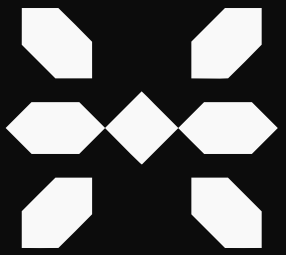
HOOX · PYNE · DOCUMENTATION



AUDITORS · CONTRIBUTORS · PARITY HUNTERS

REFERENCE MANUAL

Compatibility, status, gaps, numerical validation, pine-worker

Open-source Pine Script evaluation framework

2026-11-01 DIN A4 · HTTPS://HOOX.SH/PYNE/DOCS OPEN SOURCE

CONTENTS

01 Ecosystem

02 Commercial

03 Compatibility

04 Missing Features

05 Pine V6 Surface

06 Numerical Validation

07 Roadmap

08 Contributing

ECOSYSTEM

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title: "Ecosystem" description: "PYNE, PyneTS, pyne-worker, pyne-agent-worker, AXIS, and HOOX — what each one is and is not."

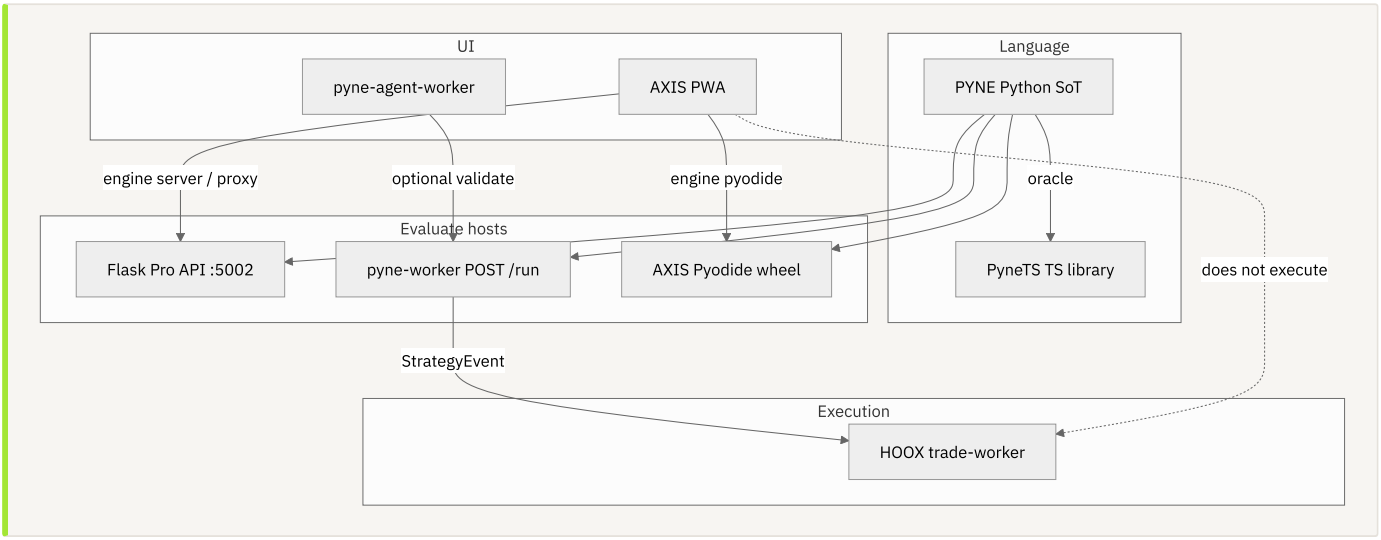
ECOSYSTEM

Abstract

The HOOX family splits **language**, **chart**, and **execution**. Mixing the names is the most common docs failure: `pyne-worker` is not `pynets`, and AXIS does not place HOOX orders.

Product	Role	Docs
PYNE (`hoox-pyne` , import `pynescript`)	Python language SoT: parse, LSP, MCP (`pyne-mcp`), Pro API, `Runtime.run`	this site
PyneTS (`@hoox-sh/pynets`)	TypeScript / Bun **library + CLI**	PyneTS
pyne-worker	Python Cloudflare® isolate — `POST /run`	pyne-worker · HOOX isolate
pyne-agent-worker	NL → Pine (Workers AI™)	agent · AXIS plugin
AXIS	Charting PWA (engines call Python)	AXIS
HOOX	Edge trade mesh	HOOX

Conceptual model



Invariant: evaluation never requires a proprietary chart host. AXIS is an optional chart. HOOX is an optional execution mesh. PyneTS is an optional TypeScript import.

Interface surface

Published versions (static snapshot; refresh manually with `bun run docs:versions` when a registry moves):

Checkouts (this repo vs sisters)

Checkout	What you open
hoox-sh/pyne	PYNE SoT (`hoox-pyne` **0.6.29**; VS Code **`hoox-sh.pyne`** **0.5.4**)
hoox-sh/pynets	Standalone `@hoox-sh/pynets` **0.5.0** (JS compile + stream)
hoox-sh/pyne-worker	Python edge `/run`
hoox-sh/pyne-agent-worker	NL authoring (**not** in this tree)
hoox-sh/axis	AXIS PWA
hoox-sh/hoox	Mesh monorepo

Evaluate contract

Flask `POST /run`, pyne-worker `POST /run`, and in-process `Runtime.run` share the evaluate contract: `script` + OHLCV + `mode` → `plots` / `events` / `alerts`. PyneTS speaks the **library** form of that envelope (`RuntimeResult`), not HTTP.

Internals

Docs are authored in product trees and synced to hoox.sh:

Tree	Public base
`pynescript/docs/pyne/**`	`/pyne/docs`
`axis/docs/**`	`/axis/docs`
`hoox/docs/**`	`/docs`

Do not author in `hoox-landing-page/content/*` — that is a sync artifact.

Invariants & edge cases

1. **Python wins** language semantics. PyneTS does not invent TradingView® behaviour.
2. **AXIS ≠ engine.** AXIS never embeds a closed interpreter; engines are plugins.
3. **AXIS ≠ execution.** Drawing a strategy on AXIS does not call `trade-worker`.
4. **Alerts ≠ orders.** `alert()` webhooks are not `StrategyEvent` forwards unless you point them at the HOOX gateway yourself.
5. **Do not** `pip install pyne` or `pip install pynescript` for this project. Dist name is **`hoox-pyne`**.
6. **Do not** `npm install pynets` for this project. The published package is **`@hoox-sh/pynets`** from the standalone hoox-sh/pynets repo. Python Runtime remains the oracle.
7. **`pine-worker`** is a leftover TypeScript Cloudflare® experiment (hoox-sh/pine-worker). It is not a product evaluate host and has no pages on this site.

Worked examples

Pick a surface

You want	Use
Parse / lint / LSP on a laptop	`pip install "hoox-pyne[lsp]"`
Embed evaluate in TypeScript	`bun add @hoox-sh/pynets`
HTTP evaluate at the edge	Deploy **pyne-worker**
Upload then keep running (Flask/VPS/container)	`PYNE_RUNNER=1` — hosted runner
Chart + editor	AXIS PWA + Flask or Pyodide
Live CEX orders from strategy events	pyne-worker `TRADE_SERVICE` → HOOX
Chat → script	**pyne-agent-worker**

Failure modes

Mix-up	What goes wrong
Opening `pyne-worker/` or `pine-worker/` inside PYNE	Directories **do not exist** — clone the sister repos; TS library work is `@hoox-sh/pynets`
Expecting AXIS to flatten a position	No trade-worker hop
Pointing AXIS at pynets as an engine	No such engine. Use Flask / Pyodide / Worker proxy
`pyne run` vs `pyne execute` vs `Runtime.run` vs `pynets run`	`pyne run` is compile-only; `pyne execute` / `pynets run` call `Runtime.run` — see modes

See also

- PYNE index
- PyneTS
- pyne-worker
- pyne-agent-worker
- Modes
- Evaluate contract
- AXIS

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title: "Commercial Licence" description: "AGPL escape hatch per organisation — closed-source use and SaaS without source disclosure. Licence sales opening soon."

COMMERCIAL LICENCE

Upcoming · per organisation · lifetime · sales open soon.

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What it removes

AGPL requires	Commercial waives
Publish modifications	Keep them private
Offer Corresponding Source to SaaS users (§13)	No source offer to your users
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Scope

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- Lifetime for 0.x/1.x line (future 2.x may need new licence).
- Subsidiaries and clients need their own.
- No warranty, no support, no SLA, no API quota. Legal permission only.

Full text: `COMMERCIAL-LICENSE.txt`.

Buy

Commercial licensing is **upcoming** — not yet on sale. To register interest: sales@hoox.dev.

FAQ

Do I need it for internal use? No — internal AGPL use without distribution/hosting for third parties needs no commercial licence.

Do I need it for SaaS? Yes, if you host PYNE for third parties and don't want AGPL §13 source obligations.

Is hosted Pro API quota included? No. API call quotas (hobby/pro/team) are a separate product.

VAT? Merchant-of-record and EU VAT handling will be announced at launch.

Refund? 14-day EU digital-goods withdrawal terms will be published at launch.

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title: "Compatibility guarantee" description: "Grouped map of what PYNE implements for Pine Script™ v5/v6 — working, partial, residual, and by-design out of scope. hoox-pyne 0.6.29."

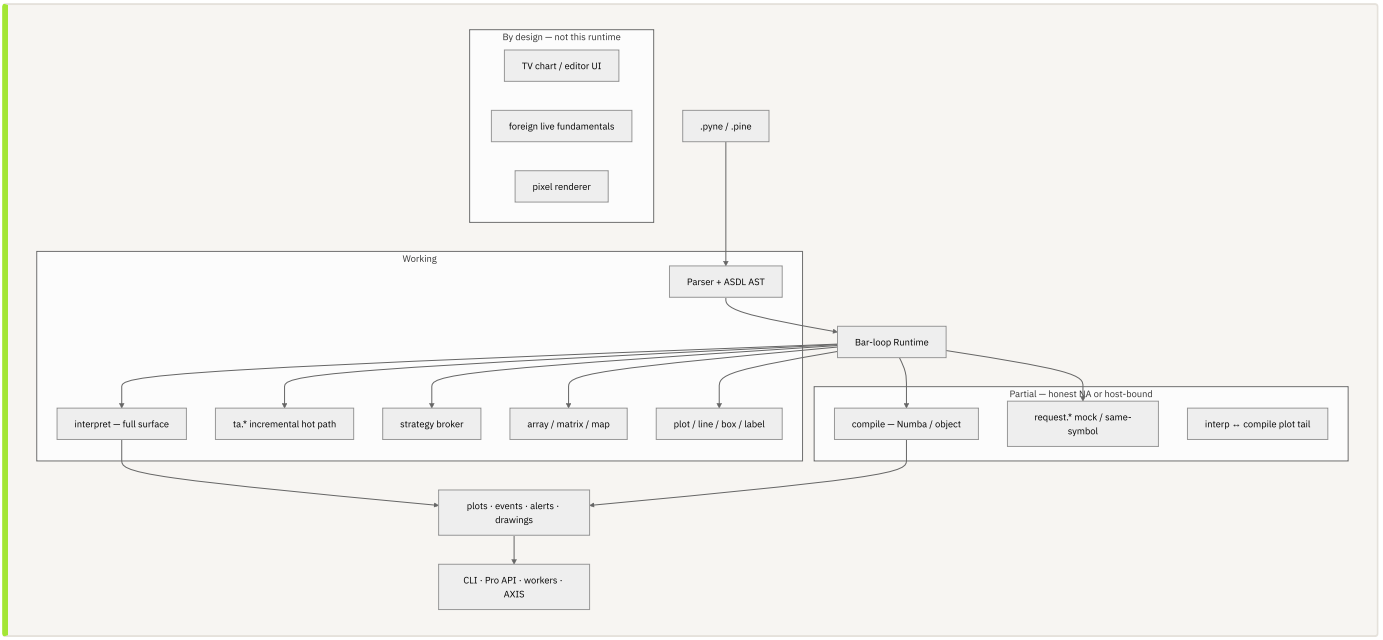
COMPATIBILITY GUARANTEE

Abstract

PYNE implements Pine Script™ **v5/v6 language core** as an inspectable pipeline: parse → AST → bar-loop evaluate. The map below is the product-facing compatibility surface for `hoox-pyne` **0.6.29**. It is **not** TradingView® platform identity (chart host, proprietary data, editor-only UI) and **not** bit-identical results vs the hosted platform on every script or bar.

Name-level dispatch lives in Pine v6 surface (`docs/pine_v6_full_surface_inventory.md`). Gaps: Missing features.

Conceptual model



Legend

Mark	Meaning
Working	Implemented, usable, covered by first-party tests
Partial	Callable but mock, host-bound, or incomplete vs hosted Pine
Residual	Known open hole on an otherwise landed surface
Out of scope	Intentionally not a TV platform clone

Dispatch snapshot (2026-10-08): **941** callables, **10** partial-heuristic, **0 missing** vs the public TV v6 function list (434 symbols). A registered name is not the same as hosted-platform semantics.

```
Language, series, TA hot path, collections, strategy broker, drawings, alerts, libraries, interpret

`request.*` feeds (honest `na`), compile eligibility, compile alerts, PyneTS vs Python oracle.

Plot MISMATCH corpus tail (P1p), `PYNE_SERIES_RING` default off.

Pixel chart, editor UX, foreign live data, bit-identical TV bars.
```

Interface surface

Corpus (set01–04 · 2477 scripts · 2026-08-09)

Open-source sets, **not shipped** in git. Not a TradingView® platform score.

Suite	Rate	Notes
Parse + unparse	**99.96%** (2476/2477)	Residual: one intentional invalid line-wrap demo
Runtime interpret (50 bars)	**100%** excl. EXPECTED_FAIL	2466 OK + 11 listed demos
set01 Runtime	**249/249**	Interpret (and prior compile sweep)

EXPECTED_FAIL are path-listed demos (`runtime.error` library guards, lower-TF security, pathological loops) — not silent suppression.

Grouped map

```
| Area | Status | Notes |
| --- | --- | --- |
| ANTLR4 grammar v5/v6 | Working | Multiline strings, soft keywords, bitwise, typed UDF returns (`F
| Parse → unparse | Working | Structural fidelity; whitespace may differ |
| `var` / `varip` / `:=` | Working | Execution-site init; start-of-bar carry |
| Functions, methods, type annotations | Working | |
| Control flow, `switch`, `once`, enums | Working | Strict bool; `na` is false in conditions. `once
| UDTs + `.new` | Working | Field defaults on compile |
| Libraries `export` / `import ns/Name/ver` | Working | In-process `register_library_source`; `Runt
| String interpolation, tuple unpack, method chaining | Working | |
| Series `[]` history | Working | OOB / negative → `na` (not wrap) |
| Line wrapping | Working | Any indent inside `(…)` and `[…]`; `table.new` is columns-then-rows (Pi
| `timeframe.change` | Working | Intraday fixed-width buckets; D/W/M on the exchange calendar (`sym



| Area | Status | Notes |
| --- | --- | --- |
| `pynescript.runtime.Runtime` | Working | Package SoT; `backend.runtime` re-exports |
| Modes `interpret` / `compile` / `auto` | Working | Defaults differ by surface — [modes](/pyne/doc
| Series caps | Working | `PYNE_SERIES_CAP` default **on** |
| Incremental `ta.*` | Working | Default **on**; `PYNE_TA_INCREMENTAL=0` full recompute |
| Derived OHLCV skip | Working | Unused `hl2`/`hlc3`/`ohlc4`/`tr` not written; `ta.vwap` still upda
| `timeout_seconds` | Working | Library / edge / Flask `/run` + `/run/batch` (omit = no budget) |
| Ring buffer | Residual | `PYNE_SERIES_RING` default **off** |



| Area | Status | Notes |
| --- | --- | --- |
| MAs | Working | `sma`/`ema`/`rma`/`wma`/`hma`/`vwma`/`kama`/`dema`/`tema`/`alma`/`sma_weighted` i
| Oscillators | Working | `rsi`/`macd`/`stoch`/`cci`/`cmo`/`tsi`/`roc`/`wpr`; incremental `aroon`/`
| Volatility / structure | Working | `atr` (Wilder `rma(tr)`), `bb`, `kc`, `stdev`, `highest`/`lowe
| Volume | Working | Incremental `obv`/`wad`/`wvad`/`cmf`/`klinger`/`mfi`/`vwap`/`nvi`/`pvi` |
| Supertrend | Working | Locked mid±factor·ATR (inc ≡ compile ≡ Numba). TV band ratchet is out of s



| Area | Status | Notes |
| --- | --- | --- |
| `array.*` | Working | Negative indices; UDT `sort_field` + `binary_search*` |
| `matrix.*` | Working | LA (`det`/`inv`/`eigen`…), predicates, `sort_field` |
| `map.*` | Working | |



| Area | Status | Notes |
| --- | --- | --- |
| `entry` / `close` / `cancel` / events | Working | `StrategyEvent` stream |
| `exit` stop/limit + OCA | Working | OHLC path, not tick path |
| `profit` / `loss` | Working | Ticks × mintick from entry avg |
| `from_entry` / `qty_percent` | Working | Interpret + compile |
| Trail | Working | `trail_offset` / `trail_points` / `trail_price` on OHLC high/low (dual-host). T
| Risk cascade | Working | `allow_entry_in`, max size/drawdown/cons-loss-days/intraday |
| Open/closed trade fields + MAE/MFE | Working | Extremes from bar high/low |
| Pending-fill average (pyramiding ≤ 0) | Working | F2 |
| Next-bar market fills (reference Pine) | Working | Market orders placed on bar N fill at the oper
```

```
| Licensed broker / exchange fills | Out of scope | In-process model only |



| Area | Status | Notes |
| --- | --- | --- |
| `plot` / `plotshape` / `plotchar` / `plotarrow` / `hline` / `bgcolor` / `barcolor` / `fill` / `pl
| `line` / `box` / `label` / `table` / `polyline` / `linefill` | Working | Coordinates snapshotted
| Drawing export parity | Working | Both hosts emit cell style/width/height/tooltip, `force_overlay
| `max_*_count` GC | Working | Interpret registry |
| `force_overlay` | Working | Export payload |
| Pixel paint | Out of scope | AXIS / clients consume series + drawings |
| Compile `hline`/`fill`/`bgcolor`/`plotshape` keys | Working | First-party key sets match interpre



| Area | Status | Notes |
| --- | --- | --- |
| Same-symbol simple OHLCV / HTF | Working | Allowlisted `ta.sma`/`ema`/`rsi`/`atr`/`wma`/`rma` on
| Foreign / complex `request.security` | Partial | Resolves to **`na`** (no invented chart-close) |
| `request.footprint` / financial / economic | Partial | Mock / feed-scaled; not a live vendor |
| `input.*` including `active` / `enum` | Working | Values + metadata |
| Bid/ask omitted | Partial | `na` when host does not supply |
| Real multi-symbol market data (B1) | Out of scope | Needs a host adapter |



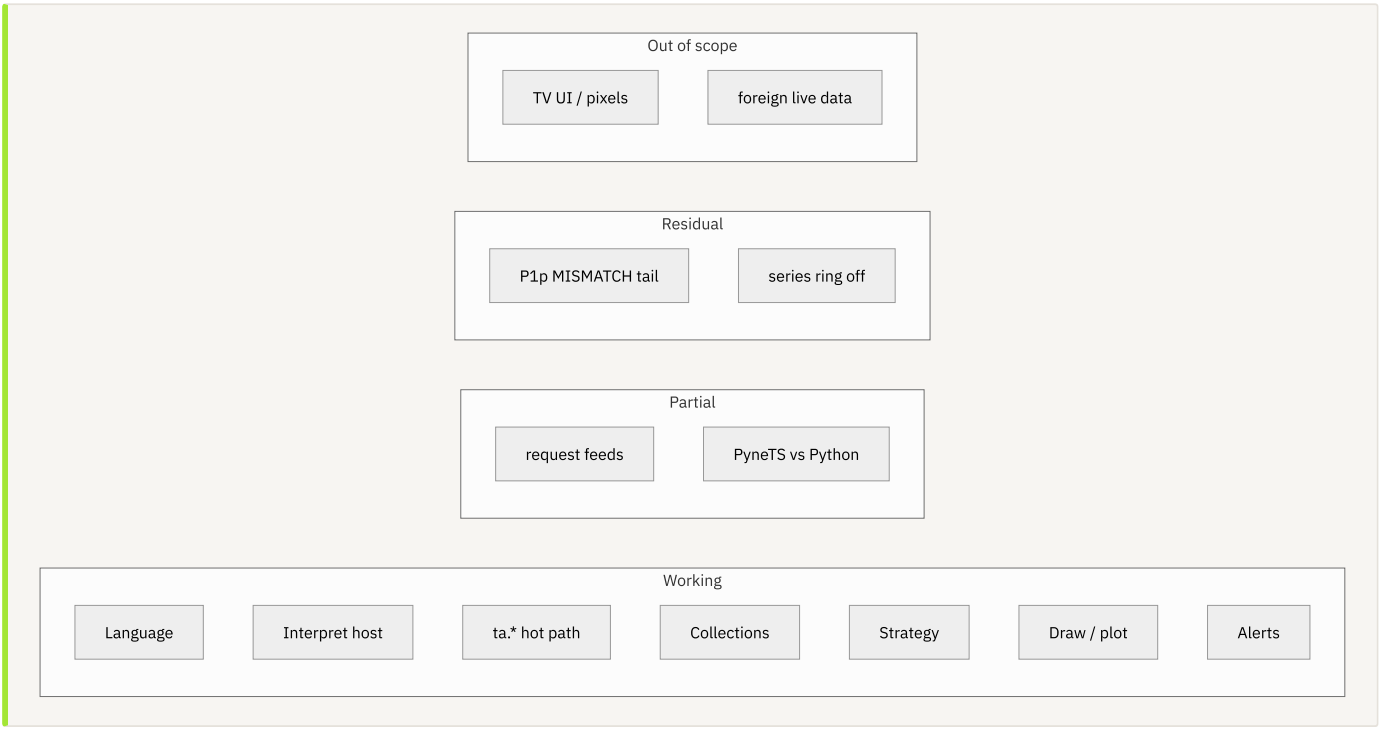
| Area | Status | Notes |
| --- | --- | --- |
| `alert()` / `alertcondition()` | Working | Frequency rules; `/run` export |
| L2 webhooks | Working | `webhook_url` / `ALERT_WEBHOOK_URL` |
| Compile alerts | Partial | Interpret is the alert oracle |
| Numba + object-mode compile | Working | `mode=auto` falls back; `import` / `request.*` / `inputs=
| Warm compile / disk IR | Working | Prewarm CLI/API; cache meta v18 |
| Interp ↔ compile plot parity | Working | Harness + first-party goldens + official TV builtins **0
| CLI / LSP / Pro API / Docker | Working | `pyne` / `pyne-lsp`; Flask `/run` |
| Flask `/run` `libraries` | Working | Max 32; also `/run/batch` |
| Flask `timeout_seconds` | Working | Optional on `/run` and `/run/batch`; omit / `≤ 0` = no timeou
| `timeframe.change` | Working | Intraday fixed-width; D/W/M exchange calendar. Interpret + compile
| `POST /optimize` / `pyne optimize` | Working | Strategy `input.*` search — not a Pine builtin |



| Surface | Status | Notes |
| --- | --- | --- |
| Python Runtime | Working | Language oracle |
| PyneTS `@hoox-sh/pynets` 0.5.0 | Partial | Interpret + JS compile on npm; Python Runtime is the c
| pyne-worker | Working | Thin Python CF host; vendors engine (may lag PyPI). Docs: [pyne-worker](/
| pyne-agent-worker | Working | NL authoring; optional `/run` validate. Docs: [agent](/pyne/docs/ag
```

Residual and out of scope (compact)

Item	Class
Interpret ↔ compile value `MISMATCH` (optional set0x community corpus)	Residual (P1p official builtins closed)
`PYNE_SERIES_RING` default-on	Residual (flagged off)
Pixel chart / editor word-wrap / TV UI	Out of scope
Foreign live fundamentals without a feed	Out of scope
TV Supertrend band ratchet	Out of scope
Tick-path trail / broker	Out of scope
Bit-identical every recursive smoother vs TV	Out of scope
Parallel bars / whole-script vectorization	Out of scope



Dual-host plot parity

Same script + OHLCV under `mode="interpret"` and `mode="compile"`; series compared with nan-aware allclose (`rtol=1e-5`, `atol=1e-6`).

Artifact	Role
`scripts/compare_interp_compile.py`	Corpus harness → `.cache/interp_compile_parity.json`
`tests/test_interp_compile_parity.py`	Always-on smoke
`tests/test_first_party_ta_goldens.py`	ATR / Supertrend / Keltner dual-host
`scripts/compare_tv_goldens.py`	Vendored TradingView Pine Logs exports — **112 / 112 columns**

Known **contract** differences (not silent value bugs):

1. Foreign / complex `request.*` → `na` on both hosts when no feed.

2. First-party `hline` / `fill` / `bgcolor` / `plotshape` keys match; harness `--ignore-hline-keys` / `--ignore-fill-keys` remain optional for leftover corpus noise.
3. Pivot-starved Auto Fib → matching `runtime.error` (`both_error_same`), not invented pivots.

Internals

Artifact	Role
`docs/compatibility_guarantee.md`	Historical long-form; this page is the map
`COMPATIBILITY.md`	Repo-root distillation of this map
`docs/pine_v6_full_surface_inventory.md`	Name-level dispatch tables
`docs/known_divergences.md`	Semantic deltas vs public Pine docs
`tests/fixtures/parity/`	First-party strategy fixtures

Invariants & edge cases

1. **Parse success ≠ evaluate success.** Round-trip can still depend on `request.*` data.
2. **Dispatch success ≠ hosted semantics.** 0 missing vs the public function list does not mean TV-identical bars.
3. **Interpret is the oracle.** Compile matches where eligible; `mode=auto` falls back.
4. **Plots are not a renderer.** Registry + series export; pixels are AXIS / clients.
5. **Floating-point is statistical** for some recursive smoothers — Numerical validation.
6. **Libraries** need in-process registration; there is no TV CDN.

Worked examples

Round-trip

```
from pynescript.ast.helper import parse, unparse

src = open("strategy.pine").read()
assert unparse(parse(src))  # structural; whitespace may differ
```

Interpret ↔ compile

```
python scripts/compare_interp_compile.py --bars 1000 --limit 50
python scripts/compare_interp_compile.py --ignore-hline-keys --ignore-fill-keys
pytest tests/test_interp_compile_parity.py -q
```

Failure modes

Observation	Read as
Parse error on a new TV release-notes feature	Surface gap — Missing features
Numerical drift on a custom smoother	Bar-mode vs list-mode; `na`; known divergences
Interpret vs compile series mismatch	Foreign `request.*`, UDF last-assign `na`, leftover corpus keys
Strategy equity ≠ TV	Commission / slippage / OHLC vs tick path
`import` unresolved	Library not in `libraries=` / registry
`/run` 400 `UNKNOWN_FIELDS`	Extra key not in `RUN_SCHEMA`

See also

- Missing features
- Pine v6 surface inventory
- Numerical validation
- Runtime modes
- Roadmap

MISSING FEATURES

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title: "Missing features" description: "Known gaps and closed Pine Script™ v6 items — what is implemented, what is partial by design, and what is out of scope. hoox-pyne 0.6.29."

MISSING FEATURES

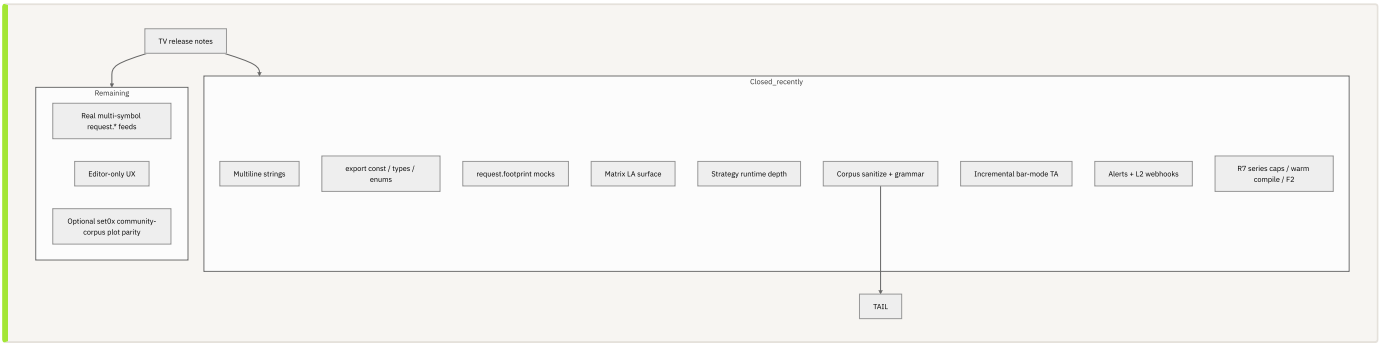
Abstract

"Missing" in PYNE is a **moving frontier**. Core v6 support is ~99%+ for core language, with post-launch TV items closed (multiline strings, library `export const`, footprint mocks, risk enforcement, matrix LA, compile-mode strategy, drawing GC, bottom-panel table fields, square-bracket wrapping) plus open-source corpus parse/Runtime hardening, incremental bar-mode TA, alert engine + L2 webhooks, series caps, warm-compile, pending-fill averaging, `timeframe.change` (intraday buckets + D/W/M exchange calendar), package Runtime SoT

(`pynescript.runtime` ; `backend.runtime` is a shim), compile object-mode residual recovery (UDF locals, nopython `None` /unicode, matrix handles, UDT returns, `x = switch` , drawing copy, `int(na)` , `round()` extras, import stubs, color type dispatch, nested UDT methods), type-aware intraday risk enforcement on both brokers, intrabar `var` rollback on realtime ticks, `request.security` barmerge on HTF resample paths, linter legacy-rule retirements, and drawing export parity (table/box/label/polyline style fields on both hosts). 0.6.10–0.6.29: engine-owned live plane (WS `/ws/live`), refcount-pooled shared sessions (F-29), opt-in delta envelopes, gevent WebSocket worker (F-30), CCXT provider caching + tick-walk sharing, live session persistence (SQLite journal), production profile + `PYNE_DATA_DIR` , Docker prod on VPS, Redis unpublished + opt-in auth, CORS for AXIS authenticated requests, separate VS Code syntax highlighting vs workbench theme (0.6.28), strategy broker next-bar market fills + parameter-only inlays + VSIX **0.5.4** (0.6.29). Remaining gaps cluster into optional set0x community-corpus plot parity (P1p official builtins closed), by-design platform differences, and editor-only features.

Authoritative narrative: `docs/missing_features.md` . Canonical IDs: `docs/ROADMAP.md` .

Conceptual model



Interface surface

Recently completed (do not re-open as "missing")

Summarized — full prose in source doc:

Area	Notes
Multiline `"""` / `'''`	Lexer + LexerBase + unparser
Library `export const` + types/enums	Parse/AST/runtime registry
UDT `sort_field` on array/matrix sort	Implemented
UDT `sort_field` on `array.binary_search*`	Implemented (interpret + compile)
`once` conditional structure	Implemented — parse/unparse, interpret (confirmed-bar latch + realtime re-fire), compile historical flags. Statement-only.
`input.*.active`	Metadata-driven
Dynamic `request.*`	Shared resolvers; mock/feed scaling
Enums runtime + input.enum	Type kind + LSP partial
Strategy open/closed trades, risk, OCA, commission	Golden tests
Drawing `*.all`, plot registry, **`max_*_count` GC**	Effects recorded; GC on package + Pro + AXIS
Numba compile + object mode + **warm-compile (H2)**	Disk IR cache, prewarm API/CLI, deploy defaults
Compile object-mode residuals	UDF locals / free-series, nopython `None` / `timestamp`, matrix handles, UDT returns, `x = switch`, drawing copy, `pine_int(na)`, sanitize chrome, `round()` extras, import stubs as `na`, color type dispatch, nested UDT methods
Official matrix LA and predicates	det/inv/eigen/...
0 missing vs public TV v6 function reference list (434 symbols)	
Runtime `_pine_defs_locked` + append-only `current_series`	Host hygiene (backend + pyne-worker)
Incremental `ta.*` (MAs, oscillators, `bb` / `kama` / `stochrsi`, volume `obv` / `wad` / `cmf` / `klinger` / `nvi` / `pvi`)	Bar mode; `PYNE_TA_INCREMENTAL=0` to disable
Series caps (T1)	`PYNE_SERIES_CAP` default ON + goldens
Pending-fill averaging (F2)	Pyramiding ≤ 0; interpret + compile broker goldens
Alert engine + **L2 webhooks (L2)**	Dual-host `/run` export, `ALERT_WEBHOOK_URL` / `webhook_url`
`request.security` foreign-na policy	Foreign/complex → `na` both backends; same-symbol OHLCV passthrough
Plot parity harness (P1p)	`scripts/compare_interp_compile.py`, first-party goldens + official TV builtins **0 MISMATCH**
`fill()` export for AXIS	series + `plot_meta` / compile `__drawings`
Square-bracket wrapping	Any-indent line wrapping inside `[…]`; `table.new` columns-then-rows
Drawing export parity	`table.cell` style storage + reference positional order; cell style/width/height/tooltip, `force_overlay`, `merged_cells` both hosts; box/label text extras; polyline curve/fill; `text.align_*` constants

Still incomplete / by design

Item	Status posture
Real (non-mock) multi-symbol `request.*` market data (**B1**)	By design for library core; feeds optional. **Foreign-na policy landed** (no invent).
`request.security` foreign data	**Not filled with real series** — emits `na` for foreign/complex (honest). Same-symbol simple OHLCV may lower.
Auto Fib / pivot-heavy scripts	Need real pivot structure (or registered ZigZag lib). Flat synthetic bars → intentional `runtime.error` (insufficient pivots); both hosts should agree (`both_error_same` in parity harness).
Pixel chart host	External (AXIS / clients)
Some unlimited-history / trim edge cases	High-level support; exotic TV behaviors may differ
Editor word-wrap / UI-only release notes	N/A
PyneTS full builtin parity (**L3**)	interpret + JS compile landed in `@hoox-sh/pynets` 0.5.0; Python remains oracle. Open rows: PyneTS missing features
Bit-identical every recursive smoother vs TV	Numerical bounds, not bits
Dual Runtime host copies (**H1**)	**Largely done** — package `pynescript.runtime` SoT + `backend.*` shims + pyne-worker thin wrap. Residual = worker-only extras (logs/profile, CF first-plot), not a forked bar loop
Warm-compile product path (**H2**)	SLOs / prewarm / deploy IR cache defaults
Runtime residual (**C1**)	set01-04 Runtime interpret **100%** excl. EXPECTED_FAIL; third-party corpus not shipped
Series cap (**T1**)	`PYNE_SERIES_CAP` default ON
Residual TA inc (**T2**)	R7 bb/kama/cmo/stochrsi + wma/hma/linreg; volume `obv` / `wad` / `cmf` / `klinger` ; `nvi` / `pvi` ; `aroon` / `dpo` / `donchian` / `kst` ; `ichimoku` / `apo` / `emv` ; `skewness` / `kurtosis` / `beta` ; `r_squared` / `comovement` ; `volume_weighted_momentum` ; `rci` / `bb_pct` / `atr_normalized` ; `cumulative_delta` / `volume_momentum` ; `sma_weighted` ; `range` / `max` / `min` ; `stoch_smooth` ; `rsi_divergence` ; `fractal` ; `engulfing` / `hammer` ; `double_top_bottom` ; `gap_detector` ; `inside_bar_pattern` ; `ema_cross_signal` ; `rsi_oversold_overbought` ; `zigzag` landed (other T2 residuals remain)
Pending-fill (**F2**)	R7 interpret + compile goldens
Plot parity residual (**P1p**)	Harness + first-party goldens + official TV builtins **0 MISMATCH**; optional set0x community corpus unmeasured
Compile leftover library stubs	Some `float(None)` / UDF arity on missing `import` libraries; INV expected-fail fixtures stay fail
TV Wilder RMA-ATR / supertrend (**F1**)	interpret ATR is `ta.rma(ta.tr)` ; Supertrend mid±factor·ATR locked. TV ratchet is out of scope
Alert webhooks (**L2**)	Pro API + pyne-worker; see Alerts

Post-v6 launch themes (historical gap list)

v6 launch items (dynamic requests, strict bool, enums, polylines, `log.*`, negative indices, truediv, ...) are largely supported or intentionally stubbed where platform-bound.

2025-2026 monthly TV updates introduced footprint types, `active` inputs, multiline strings, `export const`, UDT sort fields — tracked and largely closed.

Corpus + performance snapshot

Open-source set01-04 (**2477** scripts; measured locally, **not shipped** in-repo):

Metric	Value
Parse + unparse	**99.96%** (2476 / 2477) — residual: 1 intentional invalid line-wrap docs demo
Runtime interpret	**100%** excl. EXPECTED_FAIL — 2466 OK + **11** intentional demos
set01 Runtime	**249 / 249 (100%)**
EXPECTED_FAIL class	library `runtime.error` demos, lower-TF security guards, pathological loops, truncated mid-call scrapes
`ta_sma` bar loop (≈3.2k bars)	**~9.5×** vs full recompute
`ta_atr` bar loop	**~10×**
macd+atr+rsi+sma combo	**~8.5×**

Not TradingView® platform parity. OK% excludes only path-listed intentional demos (never soft-suppress unknown failures).

Golden: `tests/test_ta_incremental.py`, `tests/test_series_cap.py`.

Internals

Path	Role
`docs/missing_features.md`	Living gap analysis
`tests/test_v6_features.py`	v6 feature coverage
`tests/test_pine_surface_gaps.py`	Surface gap probes
`tests/test_ta_incremental.py`	Incremental TA ≡ full recompute
`tests/test_series_cap.py`	Series cap goldens (T1)
`scripts/compare_interp_compile.py`	Plot parity harness (P1p)
`docs/pine_v6_full_surface_inventory.md`	Full name-level inventory
`resource/*.g4` + builder	Grammar-level feature work
`.../technical_submodules/core.py`	`_sma_inc_update`, `_macd_inc_update`, `_atr_inc_update`, ...

Invariants & edge cases

1. **Closing a syntax gap requires grammar + builder + unparser + tests**, not only an evaluator stub.
2. **Mock data can green tests while production feeds differ** — declare data source explicitly.

3. **Compile does not fill foreign `request.security` data** — multi-asset plots stay `na` on compile unless same-symbol simple OHLCV; do not treat invent-as-close as a fix.
4. **Auto Fib needs pivots** — without swing structure (or ZigZag library registration), expect structured insufficient-pivot errors, not fib lines.
5. **"0 missing vs reference list" ≠ identical semantics for every edge case.**
6. **Perf changes must keep bar-by-bar semantics** — no whole-script vectorization; prefer golden tests vs current oracle.
7. Update this annex when `missing_features.md` changes major status lines.

Worked examples

Detect multiline string support

```
from pynescript.ast.helper import parse, unparse
src = '''//@version=6
indicator("m")
s = """a
b"""
plot(1)
'''
tree = parse(src)
assert '"""' in unparse(tree) or "'''" in unparse(tree)
```

Footprint mock call

Prefer tests in `tests/test_v6_features.py` as executable specification for `request.footprint` methods.

Incremental TA parity smoke

```
.venv/bin/python -m pytest tests/test_ta_incremental.py -q
# Disable hot path if needed:
PYNE_TA_INCREMENTAL=0 python -c "from pynescript.runtime import Runtime; ..."
```

Failure modes

Symptom	Likely gap class
Lexer error on triple quotes	Stale generated lexer (should be fixed on main)
Import member missing	Library export registration
Empty footprint rows	Mock seed / data_feed not configured
Risk not blocking entry	Old runtime path; ensure risk_kwargs applied
Runtime TIMEOUT on method-heavy scripts	Missing `_pine_defs_locked` on host
Corpus PARSE_FAIL mid-block	Truncated scrape — not a grammar hole

See also

- Pine v6 surface

- Roadmap
- Compatibility
- Technical analysis (`ta.*`)
- Alerts
- Interpret ↔ compile parity

PINE V6 SURFACE

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title: "Pine v6 surface inventory" description: "Summary of the full Pine Script™ v6 surface inventory — dispatch counts, namespaces, status schema — with pointer to the large source document. hoox-pyne 0.6.29."

PINE V6 SURFACE INVENTORY

Abstract

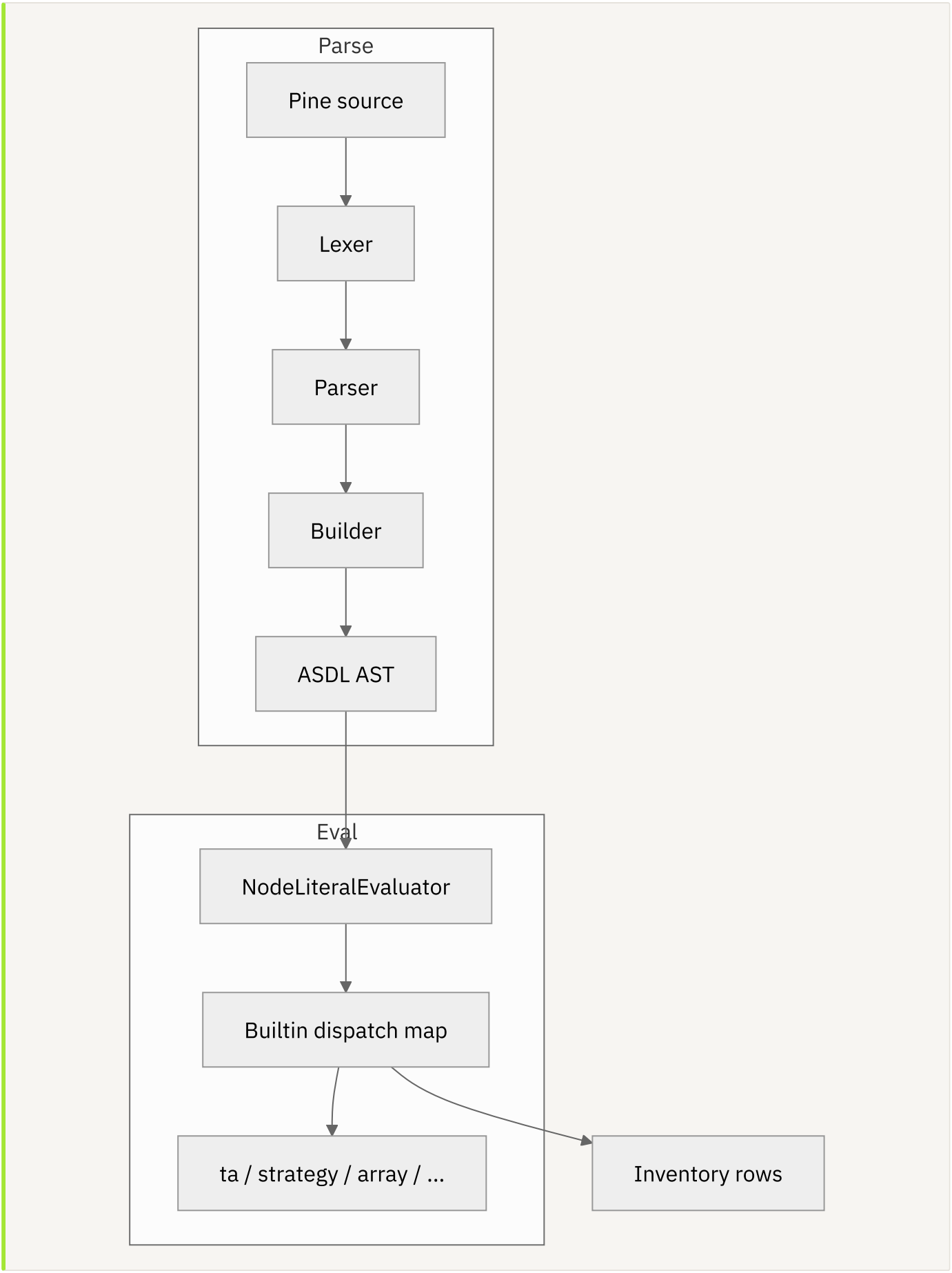
The **full surface inventory** enumerates every registered evaluator builtin, major series/variable, language construct, and known gap against Pine Script™ v6. It is deliberately large (tens of thousands of tokens when fully expanded). **This page is the navigable summary**; the exhaustive tables live in-repo at:

`docs/pine_v6_full_surface_inventory.md`

Generated snapshot referenced here: **2026-10-08**, from live `NodeLiteralEvaluator` dispatch, `builtin_metadata.json`, and design docs.

Do **not** paste the entire inventory into product docs — regenerate and link.

Conceptual model



Interface surface

Status schema

Status	Definition
Implemented	Handler/parser path exists and is usable
Partial	Mock/stub, incomplete semantics, or metadata-only
Missing	Expected on TV v6; not resolving
By design	Intentionally not full TV platform
N/A	Editor/UI-only

Each inventory row also carries: **Name**, **Namespace**, **Kind**, **Metadata** (LSP JSON present?), **Source** (e.g. `dispatch`), **Notes**.

Summary counts (2026-10-08)

Metric	Count
Dispatch builtins (callable)	**941**
Dispatch partial-heuristic (stub/mock docstring)	10
Dispatch namespaces	255
Official TV v6 function reference list	434 symbols — **0 missing** in dispatch

Top namespaces by dispatch keys

Namespace	Count
`ta`	165
`strategy`	95
`matrix`	74
`array`	68
`box`	32
`label`	29
`table`	27
`color`	26
`math`	25
`line`	24
`str`	19
`plot`	15
`timeframe`	15
`input`	14
`polyline`	14
`map`	11
`request`	11
`footprint`	9
`ticker`	9
(remaining namespaces)	smaller

Kinds covered

`function` · `series/var` · `constant` · `declaration` · `control` · `operator` · `type system` · `literal` · `semantics` · `export` · `import`

Internals

Path	Role
`docs/pine_v6_full_surface_inventory.md`	Full tables + architecture diagrams
`scripts/regenerate_inventory_summary.py`	Regeneration helper
`scripts/generate_builtin_metadata.py`	LSP metadata from live surface
Evaluator `_build_builtin_map()`	Source of dispatch truth

How to use the full file

1. Open `docs/pine_v6_full_surface_inventory.md` in the repo or Sphinx site mirror.

2. Search by namespace (`## ta` , `strategy` , ...).
3. Trust **dispatch** source rows for "is there a callable?"; read **Notes** for mock/feed caveats.
4. After large builtin PRs, regenerate summary counts before quoting them publicly.

Invariants & edge cases

1. **Inventory size is a feature** — summarization must not drop partial rows when claiming coverage.
2. **Metadata presence ≠ semantic completeness.**
3. **Partial heuristic count (10)** is docstring-based — real partials may be higher.
4. **Language constructs** (control/export) may be parser-complete without every runtime edge.

Worked examples

Quote coverage honestly

> "Against the public Pine v6 function reference (434 symbols), pynescript registered **0 missing** dispatch entries as of 2026-10-08; 941 total callables across 255 dispatch namespaces. See `docs/pine_v6_full_surface_inventory.md` ."

Programmatic check (sketch)

```
# regenerate after evaluator changes
python scripts/regenerate_inventory_summary.py
```

Failure modes

Mistake	Consequence
Pasting 134KB inventory into Mintlify page	Unreadable docs; broken UX
Citing counts without date	Stale marketing
Equating dispatch hit with TV gold	False precision claims

See also

- Missing features
- Compatibility
- Runtime builtins

NUMERICAL VALIDATION

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title: "Numerical validation" description: "Methodology and error bounds for PYNE technical and math builtins versus reference Pine Script™ implementations."

NUMERICAL VALIDATION

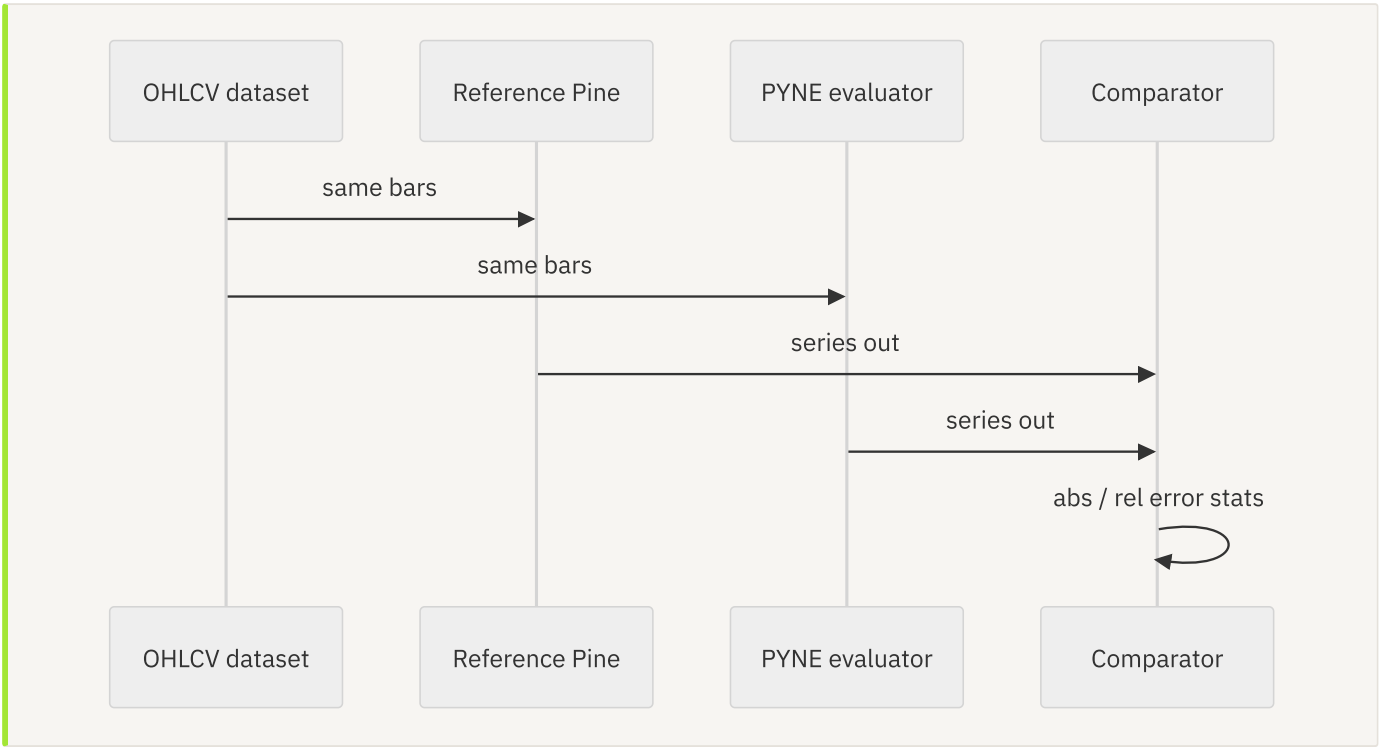
Abstract

PYNE's technical and mathematical builtins are validated for **numerical agreement** with reference Pine Script™ implementations under IEEE 754 realities. The long-form report (`docs/numerical_validation_report.md` , v1.0, Nov 2025) documents methodology, per-family error tables, and edge cases.

Headline (from that report): max observed relative error **0.0022%** (ADX, extreme volatility); average error **< 0.0005%**; deterministic ops (SMA, OBV, discrete math) often **exact**.

Treat percentages as **report-era measurements**, not eternal SLAs — re-validate after changing smoothers or bar-mode semantics.

Conceptual model



Interface surface

Acceptable thresholds (report)

Band	Relative error	Verdict
Excellent	< 0.001%	Pass
Good	< 0.01%	Pass
Acceptable	< 0.1%	Review
Unacceptable	≥ 0.1%	Fail

Metric definitions

- **Absolute error:** `|ref − pyne|`
- **Relative error:** `|ref − pyne| / |ref| × 100%` (guard zero refs)
- **Max / mean / std** over bars and scenarios

Category highlights

Family	Notes
Moving averages	SMA exact; EMA/WMA/... tiny FP error
Oscillators	RSI/stoch slightly higher smoothing error still ≪ 0.01%
Trend	ADX highest in report (0.0022% max)
Volume	OBV exact cumulative; MFI inherits typical-price FP
Stats	percentrank exact; variance/stdev excellent
Math	Discrete exact; transcendentals ~1e-15 class

Dual-host formula alignment (interpret ↔ compile)

Where fixed, these kernels **share one formula contract** across interpret (evaluator / incremental) and compile (`numba_builtins`) so plot series match bit-identical or maxdiff ≈ 0 on dual-run goldens:

Builtin	Shared contract (summary)
`ta.rsi`	Wilder: SMA seed of first `period` deltas, then RMA gain/loss; first valid at bar `period`
`ta.roc`	Standard lookback % change; `na` until lookback (no early `0.0`)
`ta.wma`	Full non-`na` window required; no partial-window reweight (nested ROC/WMA safe)
`ta.cum`	Running sum; NaNs as 0; user series (`ad = ta.cum(...)`) must not alias builtin A/D
`ta.highestbars` / `lowestbars`	Negative bars-back offset; short/all-NaN → `-1`; oldest extreme on ties

Goldens: `tests/test_compiler_numba.py` (`TestInterpCompilePlotParityFixes`) and `scripts/compare_interp_compile.py`. Residual dual-host drift (if any) tends to cluster in **nested EMA seed families** (DEMA/TEMA SMA-seed vs first-value) rather than these aligned kernels — see Compatibility.

Error distribution (report aggregate)

Error range	Share (approx.)
Exact 0	~38%
< 0.0001%	~46%
0.0001–0.001%	~14%
0.001–0.01%	~2%
≥ 0.1%	0%

Internals

Path	Role
`docs/numerical_validation_report.md`	Full tables + bias analysis
`tests/test_ta_indicators_*.py`	Executable TA tests
`tests/test_indicators.py` / builtins tests	Additional numeric guards
`tests/test_compiler_numba.py`	Interpret ↔ compile formula goldens (RSI/ROC/WMA/...)
`scripts/compare_interp_compile.py`	Corpus plot-series parity harness
Bar-mode vs list-mode (`_pine_bar_mode`)	Scalar-per-bar vs full series — compare like with like

Methodology (report)

1. Generate synthetic 1k-bar OHLCV across regimes (trend, range, vol, gaps)
2. Validate with real histories (equities, crypto, FX samples)
3. Export reference outputs; run PYNE; element-wise compare
4. Search systematic bias; inspect tails

Invariants & edge cases

1. **na propagation** must match Pine truthiness — numerical compare should mask `na` pairs.
2. **Bar-mode scalars** vs full-series list mode can look like “bugs” if misaligned.
3. **Seeded mocks** (`request.seed`) stabilize stochastic feeds for regression.
4. **Recursive smoothers** accumulate FP differently across languages; bounds > bits.
5. **Extreme prices** (near-zero, huge) tested; still within excellent band in report.

Worked examples

Relative error helper

```
def rel_err(a: float, b: float) -> float:
    if a == 0 and b == 0:
        return 0.0
    denom = abs(a) if a != 0 else abs(b)
    return abs(a - b) / denom * 100.0
```

Run TA tests

```
pytest tests/test_ta_indicators_1.py tests/test_ta_indicators_2.py -q
```

Failure modes

Symptom	Investigation
Sudden ADX drift	Check Wilder smoothing / seed bars
SMA not exact	Off-by-one window or float input casts
Good unit tests, bad TV export compare	Timezone/session alignment of bars
Only bar 0 differs	Warm-up / `na` initialization

See also

- Compatibility
- Runtime technical builtins

ROADMAP

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title: "Roadmap" description: "PYNE forward plan — dual-host residual, runtime residual, plot parity harness; drawing export parity landed. hoox-pyne 0.6.29."

ROADMAP

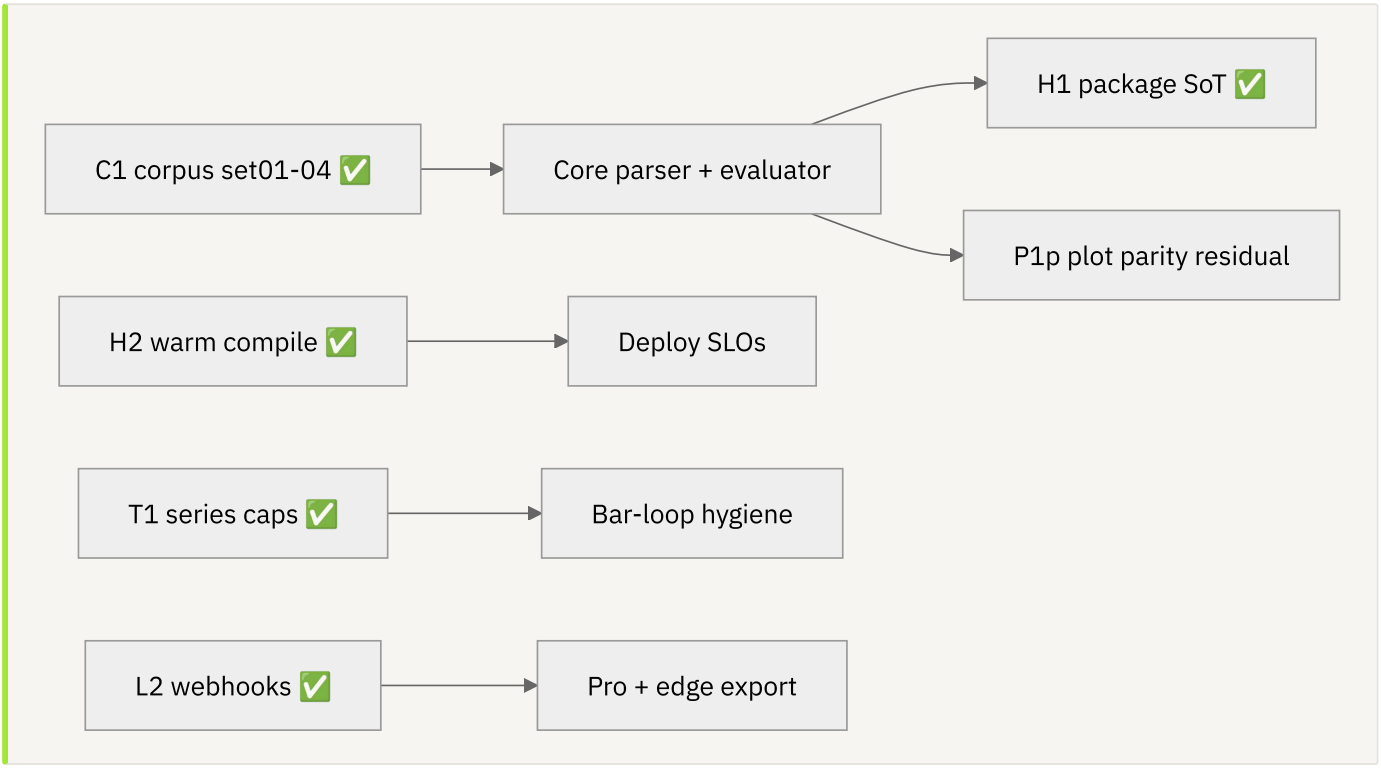
Abstract

The roadmap is a **living prioritization document**, not a contract. Source: `docs/ROADMAP.md`. Core v6 language/builtins are essentially closed. Product **warm-compile (H2)**, **series caps (T1)**, **pending-fill (F2)**, **L2 webhooks**, **corpus Runtime residual (C1)**, **package Runtime SoT (H1 largely done)** and **drawing export parity** have **landed** (set01–04 parse **99.96%**, Runtime interpret **100%** excl. intentional demos). **0.6.10–0.6.29:** engine-owned live plane (WS `/ws/live`), refcount-pooled shared sessions (F-29), opt-in delta envelopes, gevent WebSocket worker (F-30), CCXT provider caching, live session persistence (SQLite journal), production profile +

`PYNE_DATA_DIR`, Docker prod on VPS, Redis unpublished + opt-in auth, separate VS Code syntax highlighting vs workbench theme (0.6.28), strategy broker next-bar market fills + parameter-only inlays + VSIX **0.5.4** (0.6.29). What remains: **H1 worker-only extras**. **P1p** official TV builtins are **0 MISMATCH** (optional set0x community corpus unmeasured). Incremental TA and Supertrend goldens have landed.

Conceptual model

```
P0 docs honesty → P1 dual-host H1 ✅ (package SoT + shims + worker thin wrap)
                → P1 plot parity residual (P1p)
                → H2 warm compile ✅ · T1 series caps ✅ · F2 pending-fill ✅ · C1 corpus ✅
                              ↘ T2 nvi/pvi ✅ · F1 Supertrend goldens ✅
                                  → P6 long-horizon (L1 / L3; L2 ✅)
```



Interface surface

Status snapshot

Component	Status
Parser ANTLR4	v5/v6 + multiline / export const
Evaluator	incl. full var/varip, ReAssign
Builtins / TA / collections	broad (0 missing vs public TV ref list)
Strategy events + broker (OCA, commission, risk)	
Numba + object-mode compile	MVP+ (disk IR cache, `mode=auto`, `time_arr`, cache recovery; UDT/switch/matrix/drawing/round/color residuals)
Pro API + auth + Docker	(`mode` default `auto`; targets api / api-dev / lsp / **cli**)
PyPI **`hoox-pyne`**	live **0.6.29** (pypi.org/project/hoox-pyne); CLIs `pyne` / `pyne-lsp`; VS Code **`hoox-sh.pyne`** **0.5.4**
CLI surface (`check` / `format` / `compile` / `run` / `prewarm` /…)	+ Nuitka binaries + CLI Docker image
Alert engine + L2 webhooks	dual-host export + outbound webhooks
Drawing `max_*_count` GC	package + Pro API + AXIS Pyodide
Drawing export parity (table/box/label/polyline)	cell style, `force_overlay`, `merged_cells`, text extras, curve/fill, `text.align_*`; both hosts identical
Package Runtime SoT (`pynescript.runtime`)	Pro API + worker thin wrap
Strategy exits (`from_entry`, `qty_percent`, trail, risk cascade)	
`request.security` HTF simple TA + foreign-na	same-symbol OHLCV/HTF allowlist; foreign → `na`
Free-tier Pro API guards	bar/script caps, rate, concurrency, mock-only free data
Dual-host TA goldens (ATR / Supertrend / Keltner)	first-party + `test_ta_incremental` CI
Interpret dispatch + unused derived skip + volume inc	interpret dispatch inlining, unused derived OHLCV skip, incremental volume TA
Series caps (`PYNE_SERIES_CAP`)	T1
Warm-compile product path	H2 (SLOs, prewarm, deploy IR cache)
Pending-fill averaging (pyramiding ≤ 0)	F2
Interpret ↔ compile plot parity	harness + goldens landed; residual MISMATCH tail
Runtime `timeout_seconds`	wall-clock circuit breaker on interpret
UDT `array.binary_search*` `sort_field`	interpret + compile
`Runtime.run(..., libraries=)` / `POST /run libraries`	AXIS `import ns/Name/ver` support; auto fallback keeps libs
pyne-worker 0.6.0	edge `/run`; vendored engine may lag PyPI — check versions
PyneTS (`@hoox-sh/pynets`)	standalone TS library 0.5.0 — interpret + JS compile; Python remains oracle

Component	Status
pyne-agent-worker	sister NL authoring host (`POST /v1/chat`)
Full TV platform identity	out of scope
Live plane (WS `/ws/live`)	engine-owned subscription plane with delta envelopes
Shared sessions (F-29)	refcount-pooled by content hash; SQLite journal persistence
Gevent WebSocket worker (F-30)	API default worker class
Production profile	`PYNE_DATA_DIR` pinned; Docker prod on VPS; Redis unpublished + opt-in auth

Completed themes (do not re-plan as greenfield)

- Error message / typing hygiene
- `%%pinescript` Jupyter magic + sample data helpers
- `pynescript lint`
- Yahoo / AlphaVantage / CCXT + realtime datafeed wiring
- StrategyEvent emission + parity fixtures
- Colocated TS extra tool extracted. Product TypeScript library is PyneTS.
- Alert engine + dual-host export + **L2 webhooks**
- Series caps, incremental TA (bb/kama/cmo/stochrsi), parse cache, warm compile, pending-fill, drawing GC
- Interpret dispatch inlining, unused derived OHLCV skip, incremental volume TA

Open backlog (IDs stable)

ID	Item	Pri	Status
H1	Dual-host Runtime (package SoT + backend shims + pyne-worker thin wrap; residual = worker-only extras)	P1	largely done
H2	Product warm-compile (SLOs, prewarm API/CLI, IR cache on in deploy)	P1	done
C1	Corpus set01–04 residual (parse 99.96%; Runtime 100% excl. EXPECTED)	P1	done
P1p	Compile/interpret **plot parity** residual	P1	first-party + official TV builtins **0 MISMATCH**; optional set0x unmeasured
T1	Cap `current_series` to `max_bars_back` / `_SERIES_MAX`	P2	`PYNE_SERIES_CAP` default ON
T2	Incremental TA for remaining heavy kernels	P2	volume, oscillator, and band kernels
F1	ATR Wilder / TV supertrend re-baseline only with dedicated goldens	P2	mid±factor·ATR locked; TV ratchet out of scope
F2	Pending-fill averaging when pyramiding ≤ 0	P2	done
L1	v5 ↔ v6 converter maturity	P3	done — source-level rewrites both ways (semantic bool/ `na` out of scope)
L2	Webhook alerts productization	P3	Pro + pyne-worker
L3	Legacy TS Worker builtin parity	P3	sister `hoox-sh/pine-worker` only — not a product-docs target; new TS work is PyneTS
B1	Real (non-mock) `request.*` market data	—	by design; **foreign-na policy landed**

Longer horizon (not next): ML wrappers, automatic refactor, parallel/distributed eval experiments.

Priority recommendation

Horizon	Focus
Short	H1 worker extras (logs/profile); optional set0x P1p sweep
Medium	Further nested incremental TA where profiled
Long	TypeScript library work is PyneTS (**L2** done)

Landed residual notes

- **Plot parity:** Always-on smoke in `tests/test_interp_compile_parity.py`. Full corpus compare: `python scripts/compare_interp_compile.py` → `.cache/interp_compile_parity.json`. Flags `--ignore-hline-keys` / `--ignore-fill-keys` drop structural residuals.
- **Corpus residual (C1):** set01–04 parse **99.96%** (2476/2477); Runtime interpret **100%** excl. EXPECTED_FAIL (2466 OK + 11 intentional demos); set01 **249/249**. Third-party corpora are not shipped in git.
- **Foreign-na:** `request.security` on foreign/complex expressions → `na` both backends; `ChartOHLCVProvider` refuses non-chart symbols.
- **Converter (L1):** Source-level v3 → v4 → v5 → v6 upgrades and v6 → v5 downgrades via `pynescript.util.pine_convert`; exposed as `pyne convert --to 5|6` and the `pynescript.convertToV6` LSP

command/code action. Comments and string literals are never touched; type-dependent bool/`na` migration stays out of scope.

Internals

Path	Role
`docs/ROADMAP.md`	Canonical roadmap prose
`docs/PROGRESS_REPORT.md`	Historical completion narrative
`docs/COMPILER_PLAN.md`	Compile/numba plan
`scripts/compare_interp_compile.py`	Plot parity harness
`.opencode/plans/`	Engineering plans (strategy events, consolidation)
sister `hoox-sh/pyne-worker`	Python edge evaluate host — docs
sister `hoox-sh/pyne-agent-worker`	NL authoring host — docs
`@hoox-sh/pynets`	Published **0.5.0** (interpret + JS compile); Python remains oracle

Invariants & edge cases

1. **Roadmap dates lag code** — always diff against `missing_features.md` and tests.
2. **Sister repos** (pyne-worker, pyne-agent-worker) may absorb "integration" items.
3. **AXIS is a separate product tree** (`docs/axis`) — chart UX roadmap is not PYNE-core.
4. Avoid reintroducing deleted backups (`builder.py.bak`, etc.) as "plan items."
5. Do **not** re-open **H1 / H2 / T1 / T2 / F1 / F2 / L2 / C1** as greenfield — they are marked done (H1 residual = worker extras only; P1p is corpus tail) in `docs/ROADMAP.md`.

Worked examples

Track a roadmap item with a test

```
# pick a gap, write failing test first, implement, update missing_features.md
pytest tests/test_v6_features.py -k footprint -v
```

Plot parity smoke

```
pytest tests/test_interp_compile_parity.py -q
python scripts/compare_interp_compile.py   # opt-in full compare
```

TypeScript library gate (PyneTS)

```
# standalone checkout — not present as pine-worker/ in this repo
cd ../pynets && bun test && bun run typecheck
```

See PyneTS.

Failure modes

Anti-pattern	Why it hurts
Planning features already done	Duplicated work (H2/T1/F2/L2 are done)
Roadmap without tests	Unverifiable "done"
Ignoring by-design limits	Infinite platform chase

See also

- Missing features
- Interpret ↔ compile parity
- Alerts
- pyne-worker
- pyne-agent-worker
- Contributing

CONTRIBUTING

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title: "Contributing" description: "Hard constraints, workflow, and style rules for contributors and agents working on PYNE."

CONTRIBUTING

Abstract

Contributions to PYNE are welcome when they respect the repo's **mechanical invariants**: generated code boundaries, src-layout packaging, dual console scripts, test corpus costs, and encryption secrets. This page distills `CONTRIBUTING.md` and `AGENTS.md` into an operational contract for humans and agents.

Conceptual model



Interface surface

Baseline workflow

1. Fork and branch from `main`
2. Install: `make install` or hatch envs
3. Run tests: `make test` / `hatch run test:test`
4. Implement with tests (prefer TDD for non-trivial behavior)
5. Lint/format: `make lint` / `make fmt` or `hatch run lint:style` + `lint:typing`
6. Open a PR with rationale + test evidence

Official short form also lives in root `CONTRIBUTING.md`.

Everyday commands

Goal	Command
Editable install + LSP	`make install`
Full tests	`make test`
LSP only	`make test-lsp`
Backend only	`make test-backend`
Lint / format	`make lint` / `make fmt`
Fast build sanity	`make build-check`
API	`make run`
LSP process	`make run-lsp`

Hard constraints

Grammar & generated code

- **Edit only** `src/pynescript/ast/grammar/antlr4/resource/*.g4` for grammar work
- **Never hand-edit** `…/antlr4/generated/` or `…/asdl/generated/`
- Regenerate via hatch `lint:gen-parser` / project scripts; see grammar guides under `.opencode/context/`

No stale backups

Do not recreate removed backups such as `builder.py.bak` or `evaluator/builtins/technical_refactored.py`. Live `builder.py` and `technical.py` are sole sources of truth.

Future annotations

Every new Python file must start with:

```
from __future__ import annotations
```

Enforced by ruff isort `required-imports`.

Console scripts

Preferred	Alias	Role
`pyne`	`pynescript`	Click CLI
`pyne-lsp`	`pynescript-lsp`	Language Server (pygls)

Import package remains `pynescript`. Do not conflate CLI and LSP entrypoints in docs, packaging, or process managers.

Optional example-script fixture

`tests/conftest.py` only expands `pinescript_filepath` when `--example-scripts-dir` points at a local directory of `*.pine` files. **No third-party corpus is shipped** in the repository.

`tests/data/library/` is reference material, not auto-parametrized.

Builtin metadata

`builtin_metadata.json` is **generated from code**. After adding builtins:

```
python scripts/generate_builtin_metadata.py
```

Do not hand-maintain the catalog long-term.

Interpret / compile parity

When changing `pynescript.runtime.Runtime`, the compiler, or plot/series export, check **series parity** between `mode="interpret"` and `mode="compile"`:

```
# Local corpus harness (default 50 scripts × 1000 bars)
python scripts/compare_interp_compile.py --bars 1000 --limit 50

# Broader pass from a path list; ignore one-sided hline/fill keys
python scripts/compare_interp_compile.py --file-list path.txt --limit 0 --workers 4 --timeout-sec 3
```

Always-on unit coverage lives in `tests/test_interp_compile_parity.py` (harness helpers + a small smoke subset). Optional longer path:

```
pytest tests/test_interp_compile_parity.py -m interp_compile_full
# or: PYNE_INTERP_COMPILE_FULL=1 pytest tests/test_interp_compile_parity.py
```

Report artifact: `.cache/interp_compile_parity.json`. Value mismatches on shared `series` keys are regressions. First-party `hline` / `fill` / `bgcolor` / `plotshape` / `barcolor` / `plotarrow` / `plotbar` / `plotcandle` keys match (0.3.12); `drawings` is geometry-only. `--ignore-hline-keys` / `--ignore-fill-keys` are optional for leftover

corpus noise. End-user mode semantics: Evaluate scripts — Runtime modes.

Secrets & crypto

- `scripts/build/.metadata.key` is **gitignored**
- CI must supply a stable `CRYPTO_KEY` (or `METADATA_KEY`) for reproducible `.enc` blobs
- Never commit Fernet keys or API admin tokens

Internals (where to change what)

Want to...	Look at
Parse / unparse	`src/pynescript/ast/helper.py`
Add builtin	`src/pynescript/ast/evaluator/builtins/<ns>.py` + metadata generator
LSP feature	`src/pynescript/langserver/features/` + `server.py`
CLI subcommand	`src/pynescript/__main__.py`
Grammar	`resource/*.g4` then regenerate
Pro API route	`backend/api/preview.py`, `backend/app.py`
Runtime modes / compile fallback	`src/pynescript/runtime/host.py` (`Runtime.run`); `backend/runtime.py` is a shim
Interp/compile series parity	`scripts/compare_interp_compile.py`, `tests/test_interp_compile_parity.py`
TS library	sister repo hoox-sh/pynets / `@hoox-sh/pynets`

Style

- **Ruff** line length 120; broad rule set in `pyproject.toml`
- **Black** target `py310` (via toolchain config)
- **Mypy** strict-ish; exemptions for generated grammar, `evaluator.builtins.*`, tests
- Voice for docs: academic nerdy-cool per `docs/WRITING.md` — no empty marketing adjectives

Release (this repo)

Match root `CONTRIBUTING.md` and `.github/workflows/publish.yml` / `release.yml` / `ghcr.yml`:

1. Bump `__version__` in `src/pynescript/__about__.py` and update `CHANGELOG.md`.
2. Align `vscode-extension/package.json` when shipping the VSIX together.
3. Ensure CI is green on `main`.
4. Tag and push: `git tag vX.Y.Z && git push origin vX.Y.Z`.
5. **Publish** (`publish.yml`) builds sdist/wheel (`hoox_pyne-*.whl`) and uploads `hoox-pyne` to PyPI on `v*` tags.
6. **Build & Release** (`release.yml`) attaches Nuitka CLI/LSP binaries + VSIX to the GitHub Release.
7. **GHCR** (`ghcr.yml`) publishes `ghcr.io/hoox-sh/pyne/{cli,lsp,api}:X.Y.Z` on `v*` tags.

Dry-run PyPI: Actions → **Publish** → Run workflow → `dry_run=true`. Local: `python -m build && twine check dist/*`.

AXIS charting releases live in hoox-sh/axis.

PyneTS contributions

- Work in hoox-sh/pynets, not a `pine-worker/` directory in this repo
- Bun for install / test / typecheck; `bun run generate` after PYNE `.g4` changes
- Python remains the oracle — see PyneTS parity
- Do not copy PyneTS sources into this tree; work in the standalone hoox-sh/pynets repo / `@hoox-sh/pynets` package
- Python Runtime remains the oracle.

Failure modes for contributors

Mistake	Fallout
Editing generated ANTLR/ASDL	Overwritten / review hell
Skipping corpus cost awareness	10-minute “unit” tests
Hand-editing metadata JSON	Drift from dispatch
Changing compile/export without parity check	Silent chart/series drift vs interpret
Committing `.metadata.key`	Secret leak; rotate immediately
PR without lint	CI red on ruff/mypy

See also

- Local development
- CI
- Release
- PyneTS
- pyne-worker
- pyne-agent-worker
- Roadmap
- Root `AGENTS.md`, `CONTRIBUTING.md`, `CODE_OF_CONDUCT.md`